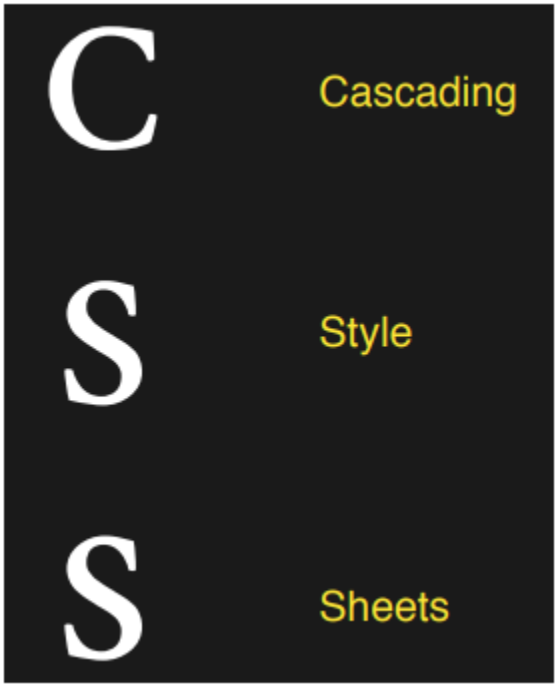
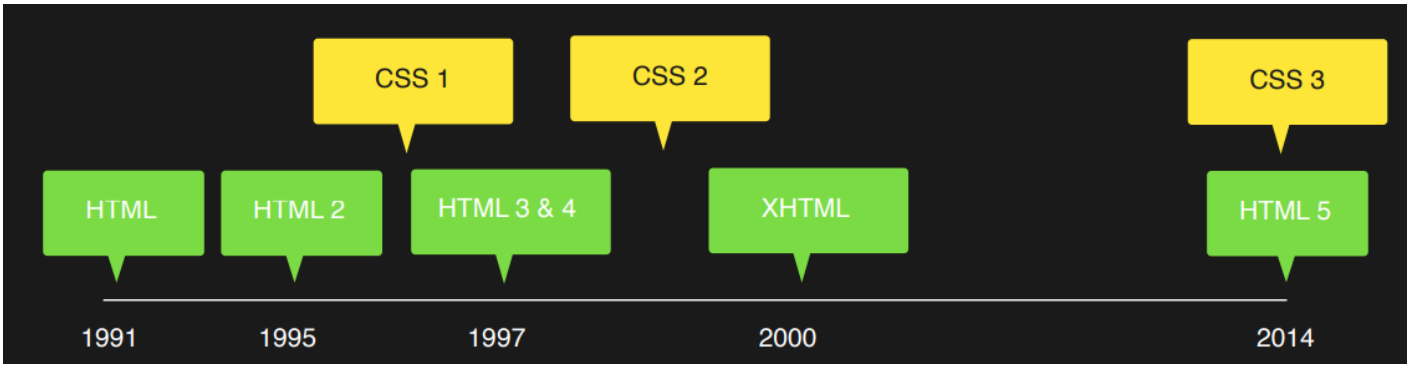
S.P.M College,Udantpuri

Bachelor Of Computer Application (BCA)

Part -1 (Paper-1) **– Hira Kumar**

Web Development



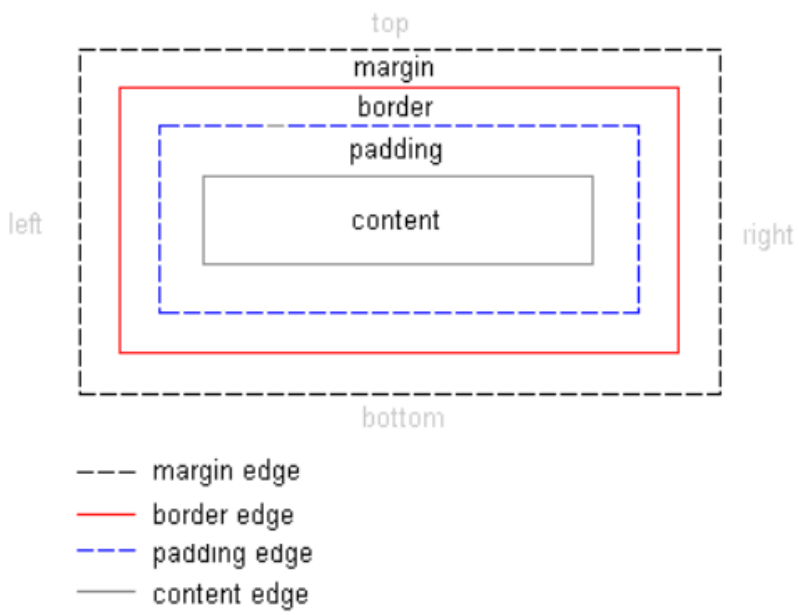


HIRA KUMAR

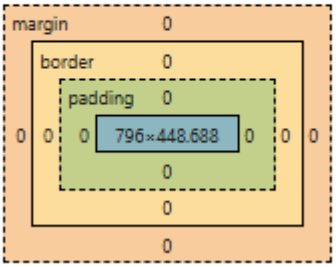
❖ CSS Box Models

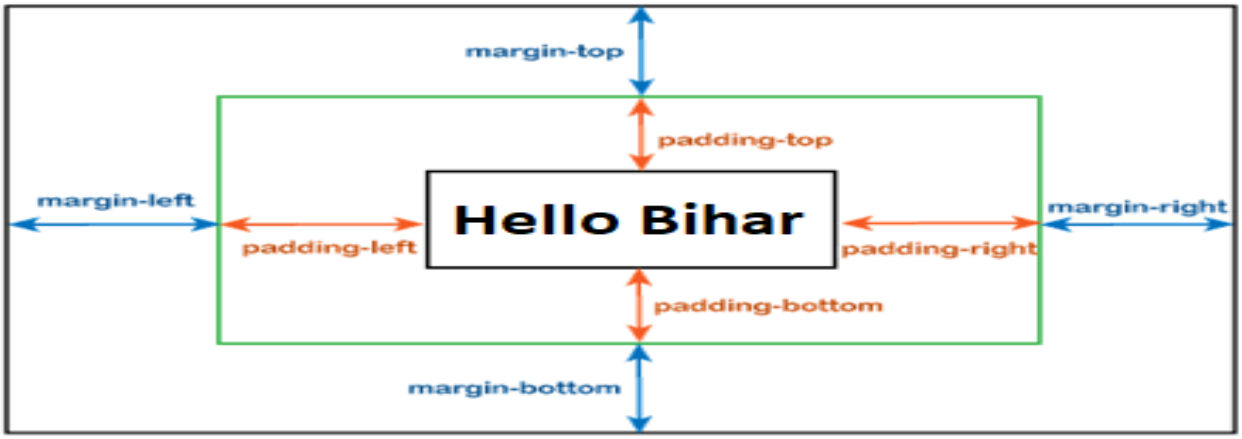
- All HTML elements can be considered as boxes. The term "box model" is used when talking about design and layout.
- The box model is essentially a box that wraps around HTML elements (h1, p,div,…), and it consists of: margins, borders, padding, and the actual content.
- The box model allows us to place a border around elements and space elements in relation to other elements.

❖ Box Model Structure



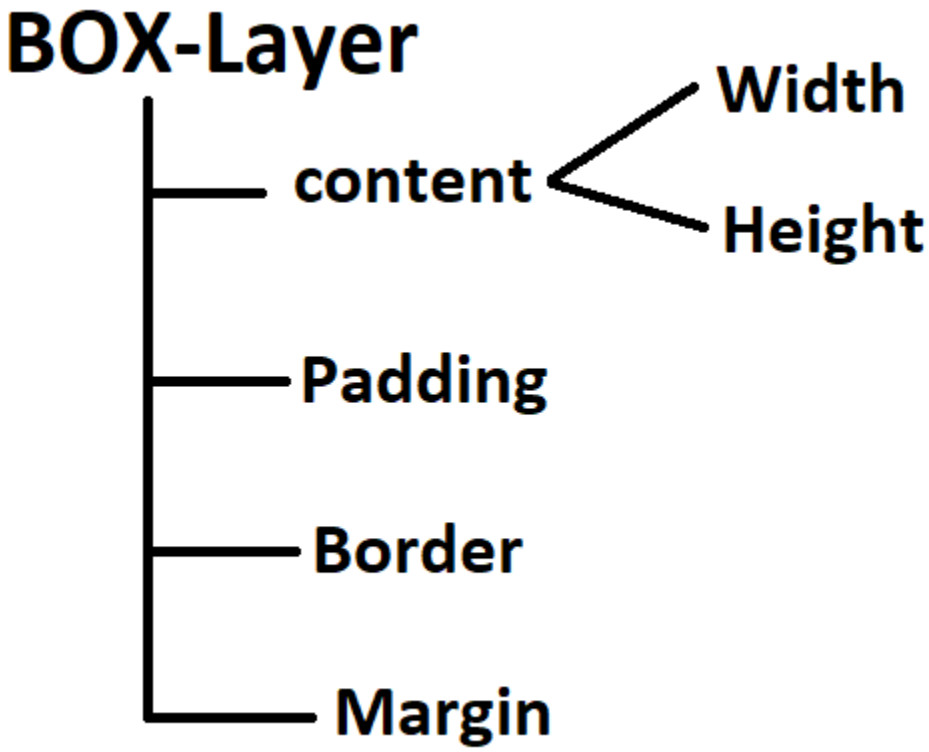
OR





❖ Box Model Properties/layer

- In Css Box model divided into 4 layers :-
    1. Content
    2. Padding
    3. Border
    4. Margin



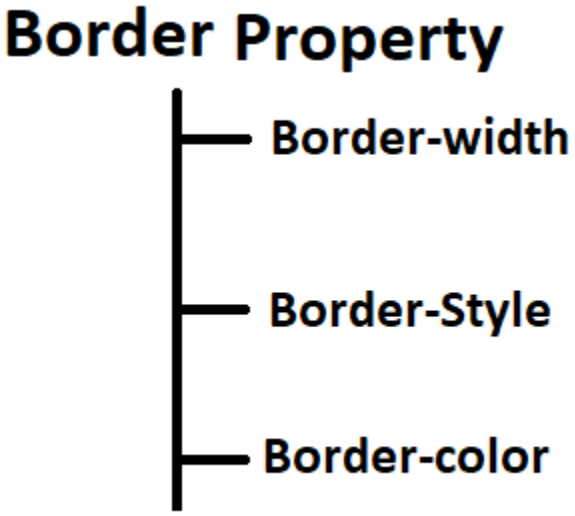
➢ **Content** - The content of the box, where text and images appear.

➢ **Padding** - Clears an area around the content. The padding is affected by the background color of the box.

- **Border** - A border that goes around the padding and content. The border is affected by the background color of the box.
- **Margin** - Clears an area around the border. The margin does not have a background color, it is completely transparent.

❖ Border Property

- The border property in CSS is used to give a border around a box / around an element.
- **The Style value is required, without it, the border won't appear.**
- Values can be in any order,but mandatory including all three is typical.



1. Border Width Property

Used to thickness of the border.

2. Border style Property

Used to different types of line of the border.

3. Border color Property

Used to color of the border.

Syntax :-

```
{
    Border-style : valaue;
    Border-width:value;
    Border-color:value;
}
```

Property	Value
Border-widht:	10px;
Border-style:	None; → No border Hidden; → same as none, but for tables Dotted; → dotted border Dashed ; → dashed border Solid; → solid border Double; → double border Groove; → 3d grooved effect Ridge;→ 3d ridge effect Insert → 3d insert effect Outset → 3d outset effect
Border-color	Red; → Named colors #ff0000; → hexadecimal rgb(255,0,0); → RGB transparent; → transparent currentcolor (current text color)

Example :-

{

```
    border-width:2px;
     border-style: solid;
    border-color:red;
}
```

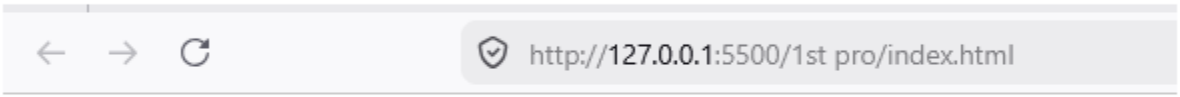
```
7   <style>
8 div{
9     background-color: blue;
10    color: white;
11    border-width:2px;
12    border-style: solid;
13    border-color: red;
14  }
15    </style>
16  </head>
17  <body>
18    <div>
19      <p> Biharsharif</p>
20    </div>
21  </body>
```

✓ Shorthand use

Syntax :→ border : width style color;

Example:→ border : 2px solid red;

```
<body>
    <a style="border : 2px solid red;"> Biharsharif</a>
</body>
```



❖ Side-Wise Border / Directional Borders

- In directional borders let you apply a border to only one specific side of an element,instead of all four sides at once.

Border-top-width:5px;

Border-top-style:solid; Border-top-color:black; **OR Shortcut** Border-top: 2px dashed blue;	
Border-top: 2px dashed blue; border-right: 3px solid green; border-bottom: 3px; border-left :2px; border-radius: 3px;	→ /* top edge only*/ → /* right edge only*/ → /* bottom edge only*/ → /* left edge only*/

```
  <style>
    a{
      background-color: red;
      color: white;
    }
  </style>
</head>
<body>
    <a style="border : 2px solid black;"> Biharsharif</a> <br><br>
    <a style="border-top : 2px solid black;"> Biharsharif</a> <br><br>
    <a style="border-bottom : 2px dashed black;"> Biharsharif</a> <br><br>
    <a style="border-right : 2px solid black;"> Biharsharif</a> <br><br>
</body>
```

❖ Border Value accept

- These accept 1 to 4 values in top-right-bottom-left order (clockwise)
- Syntax – border property : top right bottom left
- Example – border-width : 2px 1px 5px 10px;
  border-style: solid dashed dotted solid;

border-color: red green blue orange;

Value Rules :

- write only 1 value → all 4 sides
- write only 2 values → top/bottom + right/left
- write only 3 values → top + right/left + bottom
- write only 4 values → top+right+bottom+left

❖ Border-radius Property

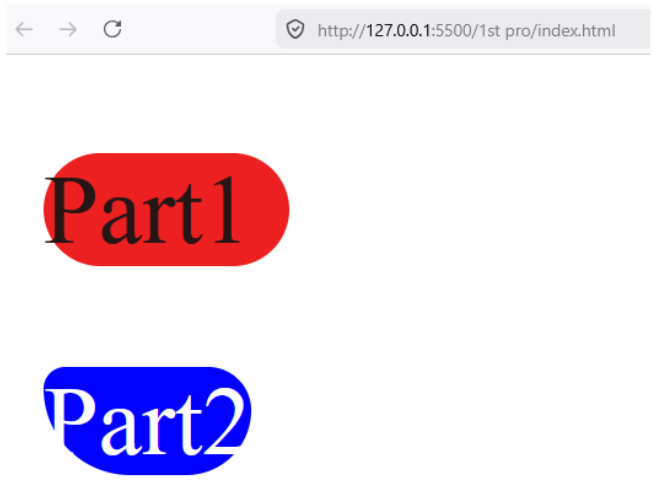
- Border-radius is a CSS property that adds rounded corners to an element's outer border edge. It "cuts off" the sharp 90-degree corners and replace them with a smooth curve.
- Syntax –

Border-radius : 10px; /* 1 value use */
Border-radius : 10px 10px; /* 2 value use */
Border-radius: 5px 10px 15px 20px; /*4 value use */

- Example –

```html
<style>
p{
    background-color: rgb(236, 32, 32);
    color: rgb(39, 21, 21);
   border-radius:100px;
   width:40px;
  }
  a{
    background-color: blue;
    color: white;
    border-radius:5px 10px 15px 20px;
  }
</style>
</head>
<body>

  <p> Part1 </p>
  <a> Part2</a>
```



❖ Padding Property

- Padding property is used to create space inside the element, between the content and the border.
- Syntax : padding: position;

  (position → top, right, bottom, left)

Property	Value	work
Padding:	10px;	All sides
Padding:	10px 20px;	Top/bottom | left/right
Padding:	10px 20px 30px;	Top | left/right | bottom
Padding:	10px 20px 30px 40px;	Top|right|bottom|left

Padding Values

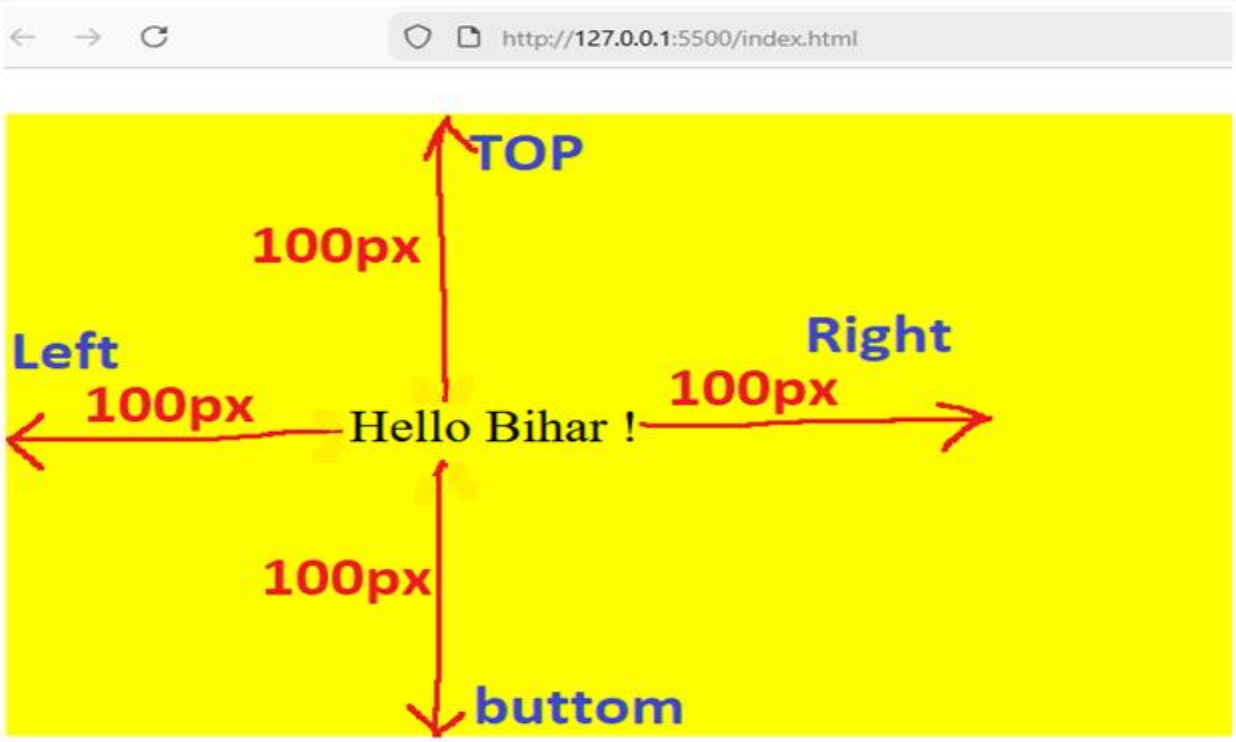
Length Values		
Property	**Value**	**Work**

Padding:	20px;	Pixels
Padding:	1em;	Relative to font-size
Padding:	1rem;	Relative to root font-size
Padding:	5%;	In precentage

Individual side properties and value	
Property	**Value**
Padding-top:	20px;
Padding-right:	10px;
Padding-bottom:	10px;
Padding-left:	25px;

```html
<title>hira css</title>
<style>
  div p{
    background-color : yellow;
   padding:100px;
  }

</style>
<body>
  <div>
    <p> Hello Bihar !</p>
    </div>
    <div>
      <a> Learn HTML & CSS</a>
    </div>
```



Learn HTML & CSS

❖ Margin Property

- Margin Property is used to create space outside the element.
- Syntax → margin: position;
  (position = top, right, bottom, left)

Property	Value	work
margin:	10px;	All sides
margin:	10px 20px;	Top/bottom | left/right
margin:	10px 20px 30px;	Top | left/right | bottom
margin:	10px 20px 30px 40px;	Top|right|bottom|left

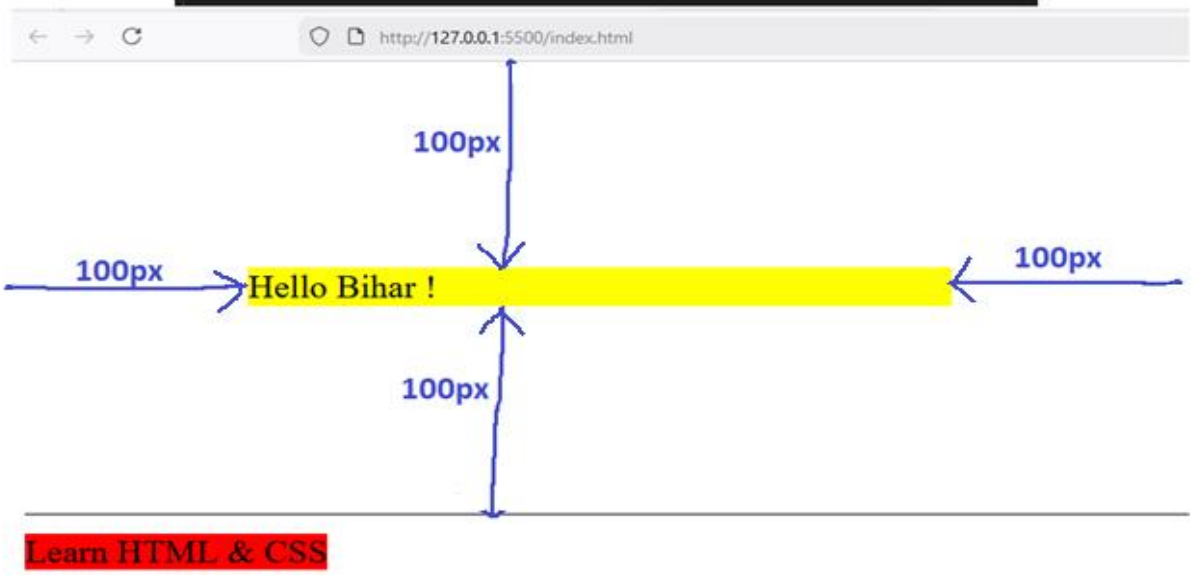
Margin Values

Length Values		
Property	**Value**	**Work**
Margin:	20px;	Pixels
Margin:	1em;	Relative to font-size
Margin:	1rem;	Relative to root font-size
Margin:	5%;	In precentage

Auto Values		
Property	**Value**	**Work**
Margin:	Auto;	Browser calculates the margin
Margin:	0 auto;	Center block elements horizontally

Individual side properties and value	
Property	**Value**
Margin-top:	20px;
Margin-right:	10px;
Margin-bottom:	10px;
Margin-left:	25px;

```html
<title>hira css</title>
<style>
  div p{
    background-color : yellow;
  margin:100px
  }
</style>
<body>
  <div>
    <p> Hello Bihar !</p>
    </div><hr>
    <div>
      <a> Learn HTML & CSS</a>
    </div>
```



❖ Width property

- Width property is used to set the width of an element.
- Syntax → width: value;

Length/keyword Values		
Property	**Value**	**Work**
Width:	Fit-content;	Shrinks to fit content
Width:	Min-content;	Smallest possible width
Width:	Max-content;	Width based on content
Width:	Inherit;	Inherits from parent
Width:	initial	Sets to default (auto)

Absolute lengths/units		
Property	**Value**	**Work**
Width:	500px;	Fixed pixel width
Width:	10in;	Inches
Width:	20cm;	Centimeters
Width:	200mm;	Millimeters
Width:	12pt;	Points (1pt = 1/72 inch)
Width:	2pc;	Picas (1pc = 12pt)

Relative lengths		
Property	**Value**	**Work**
Width:	50%;	50% of parent's width
Width:	50vw;	50% of viewport width
Width:	50vh;	50% of viewport height
Width:	25em;	Relative to elements font-size
Width:	25rem;	Relative to root font-size
Width:	50vmin;	50% of smaller viewport dimension
Width:	50vmax;	50% of larger viewport dimension
Width:	100cqw	100% of container width

Min/max & responsive image properties and values		
Property	**Value**	**Work**
Max-Width:	1200px;	Till 1200px in big screen
Min-width:	300px;	Till 300px in small screen
Width:	Clamp(200px,50%,800px);	Flexible image size

Invalid ❌

Width:500;	Unit missing
Width: 50%px;	Double unit

Width: auto 100px; Width: red; Width: calc(100%-);	Multiple values Invalid value Incomplete calc

❖ Height property

- Height property is used to set the height of an element.
- Syntax – height: value;

Example –

Keyword values

Property	Value	Work
Height:	Auto;	Default- browser calculates
Height:	Inherit;	Takes value form parent element
Height:	Initial;	Resets to default value (auto)
Height:	Unset;	Inherit if proper is inheritable
Height:	Revert;	Reverts to browser default styling
Height:	Revert-layer;	Reverts to previous cascade layer value

Length values

Absolute lengths		
Property	**Value**	**work**
Height:	100px;	Pixels
Height:	5cm;	Centimeters
Height:	5mm;	Millimeters
Height:	2in;	Inches
Height:	12pt;	Points
Height:	1pc;	Picas
Height:	10Q;	Quarter-millimeters
Relative lengths – fonts relative		

Height:	10em;	Relative to elements font size
Height:	10rem;	Relative to root elements
Height:	5ex;	Relative to height of “x” character
Height:	5ch;	Relative to width of”0” character
Height:	5Lh;	Relative to element’s line height
Relative lengths		
Height:	50%;	Relative to parent element
Height:	100vh;	100% of viewport height
Height:	50vh;	50% of viewport height
Height:	100 svh	Small viewport height
Height:	5vmin	5% of the smaller dimension